

Activity-4

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Question 1 - Empirical Cumulative Distribution Function (ECDF)

Scenario

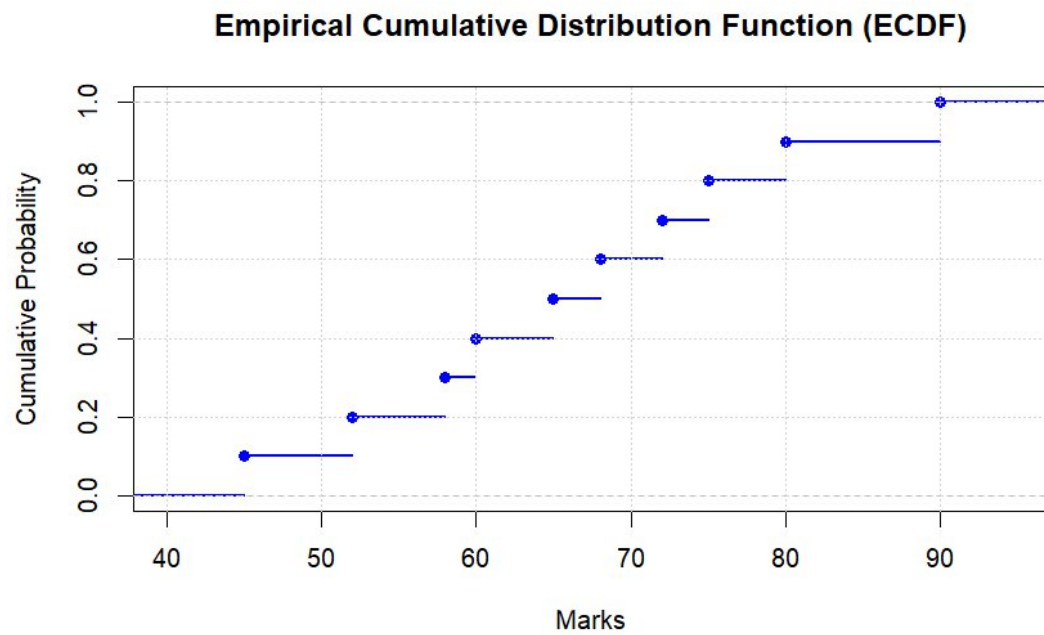
A teacher recorded the marks of students in a Data Science quiz. The teacher wants to determine the cumulative proportion of students scoring at or below a given mark.

Dataset

Student	Marks
S1	45
S2	52
S3	58
S4	60
S5	65
S6	68
S7	72
S8	75
S9	80
S10	90

Question

```
1 marks <- c(45, 52, 58, 60, 65, 68, 72, 75, 80, 90)
2 ecdf_marks <- ecdf(marks)
3 plot(ecdf_marks,
4       main = "Empirical Cumulative Distribution Function (ECDF)",
5       xlab = "Marks",
6       ylab = "Cumulative Probability",
7       col = "blue",
8       lwd = 2)
9 grid()
10 cum_prob <- ecdf_marks(marks)
11 data.frame(
12   Marks = marks,
13   Cumulative_Probability = cum_prob
14 )
```



The ECDF plot shows the cumulative proportion of students scoring at or below each mark. For example, 50% of students scored 65 marks or less, 80% scored 75 marks or less, and 100% scored 90 marks or less, indicating the complete distribution of the quiz marks.

Question 2 - Empirical Cumulative Distribution Function (ECDF)

Scenario

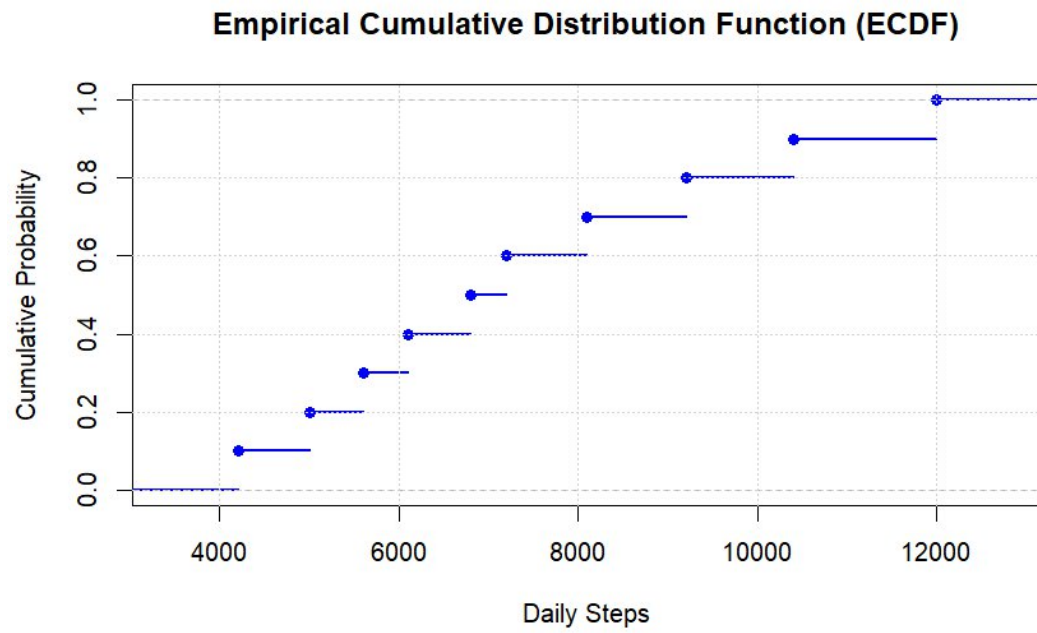
A fitness trainer recorded the number of daily steps completed by ten participants during a health challenge.

Dataset

Participant	Daily Steps
P1	4200
P2	5000
P3	5600
P4	6100
P5	6800
P6	7200
P7	8100
P8	9200
P9	10400
P10	12000

Question

```
1 steps <- c(4200, 5000, 5600, 6100, 6800, 7200, 8100, 9200, 10400, 12000)
2 ecdf_steps <- ecdf(steps)
3 plot(ecdf_steps,
4       main = "Empirical Cumulative Distribution Function (ECDF)",
5       xlab = "Daily Steps",
6       ylab = "Cumulative Probability",
7       col = "blue",
8       lwd = 2)
9
10 grid()
11 cum_prob <- ecdf_steps(steps)
12 data.frame(
13   Daily_Steps = steps,
14   Cumulative_Probability = cum_prob
15 )
16 ecdf_steps(8000)|
```



The ECDF plot shows the cumulative proportion of participants who walked at or below a given number of daily steps. It indicates that 60% of participants walked 8,000 steps or fewer, making the ECDF useful for understanding the distribution and cumulative percentages of daily step counts.

Question 3 - Highly Skewed Distribution (Histogram and Density Plot)

Scenario

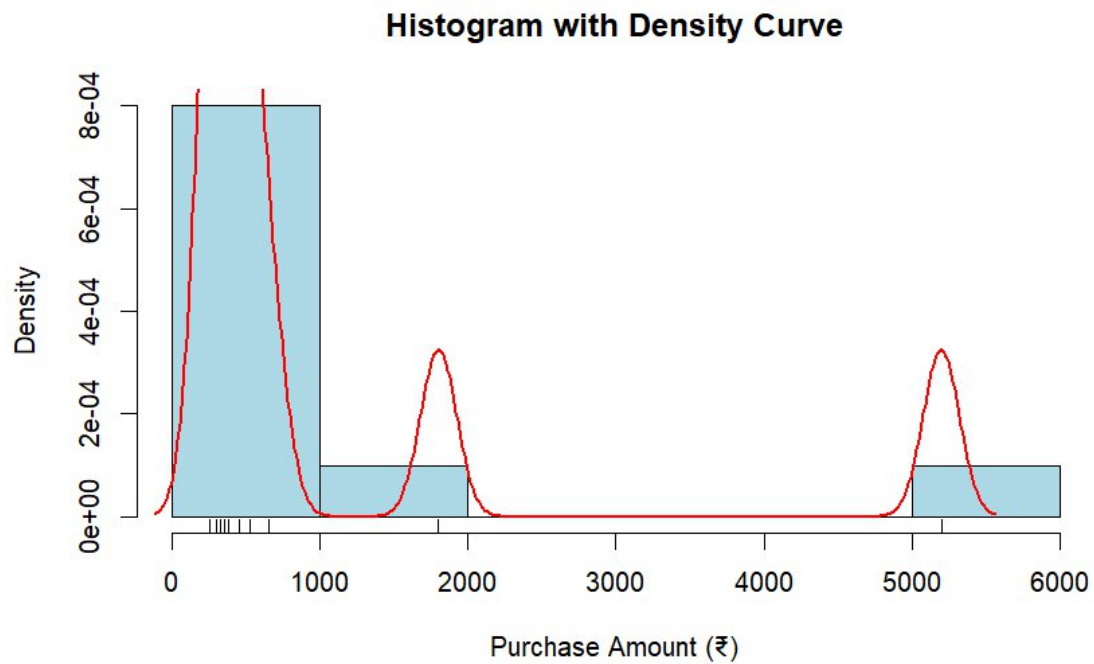
An online shopping company recorded the purchase amount (₹) made by customers during a promotional sale. Most customers purchased inexpensive items, while a few made very expensive purchases.

Dataset

Customer	Purchase Amount (₹)
C1	250
C2	300
C3	320
C4	350
C5	380
C6	450
C7	520
C8	650
C9	1800
C10	5200

Question

```
1 purchase <- c(250, 300, 320, 350, 380, 450, 520, 650, 1800, 5200)
2 hist(
3   purchase,
4   probability = TRUE,
5   col = "lightblue",
6   border = "black",
7   main = "Histogram with Density Curve",
8   xlab = "Purchase Amount (₹)"
9 )
10 lines(
11   density(purchase),
12   col = "red",
13   lwd = 2
14 )
15 rug(purchase)
```



The histogram and density plot show that the distribution is positively (right) skewed. Most purchase amounts are clustered between ₹250 and ₹650, while a few very large purchases (₹1800 and ₹5200) create a long tail on the right side of the distribution.

Question 4 - Quantile-Quantile (Q-Q) Plot

Scenario

A hospital administrator collected the waiting times (in minutes) for patients in the outpatient department and wants to verify whether the data follows a normal distribution.

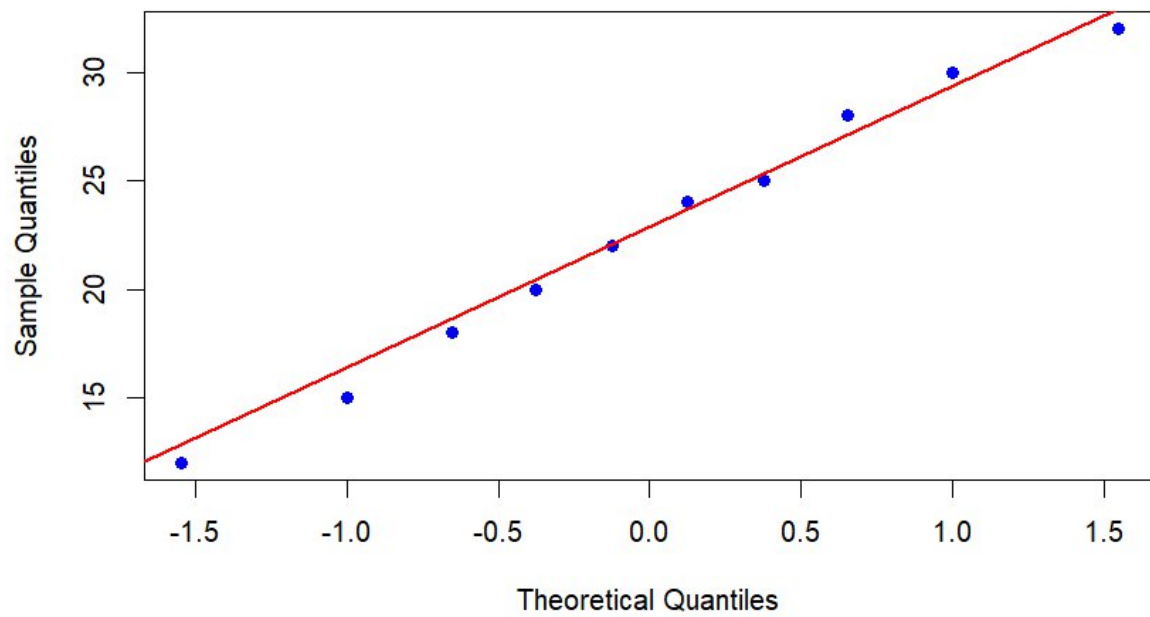
Dataset

Patient	Waiting Time (minutes)
P1	12
P2	15
P3	18
P4	20
P5	22
P6	24
P7	25
P8	28
P9	30
P10	32

Question

```
1 waiting_time <- c(12, 15, 18, 20, 22, 24, 25, 28, 30, 32)
2 qqnorm(
3   waiting_time,
4   main = "Normal Q-Q Plot of Waiting Times",
5   pch = 19,
6   col = "blue"
7 )
8 qqline(
9   waiting_time,
10  col = "red",
11  lwd = 2
12 )
```

Normal Q-Q Plot of Waiting Times



The Q-Q plot shows that the data points lie approximately along the reference line, indicating that the waiting times approximately follow a normal distribution with only minor deviations.

Question 5 - Highly Skewed Distribution and Q-Q Plot

Scenario

A bank analyzed the monthly income (₹) of ten customers applying for home loans. A few customers have exceptionally high incomes compared to the majority.

Dataset

Customer	Monthly Income (₹)
C1	18000
C2	22000
C3	24000
C4	26000
C5	28000
C6	30000
C7	34000
C8	42000
C9	85000
C10	200000

Question

```
1 income <- c(18000, 22000, 24000, 26000, 28000,
2             30000, 34000, 42000, 85000, 200000)
3 hist(
4   income,
5   col = "lightblue",
6   border = "black",
7   main = "Histogram of Monthly Income",
8   xlab = "Monthly Income (₹)"
9 )
10 qqnorm(
11   income,
12   main = "Normal Q-Q Plot of Monthly Income",
13   pch = 19,
14   col = "blue"
15 )
16 qqline(
17   income,
18   col = "red",
19   lwd = 2
20 )
```

3. The income data is highly positively (right) skewed and does not follow a normal distribution.

4. The Q-Q plot supports this conclusion because the points deviate significantly from the reference line, especially in the upper tail due to the high-income outliers (₹85,000 and ₹200,000).